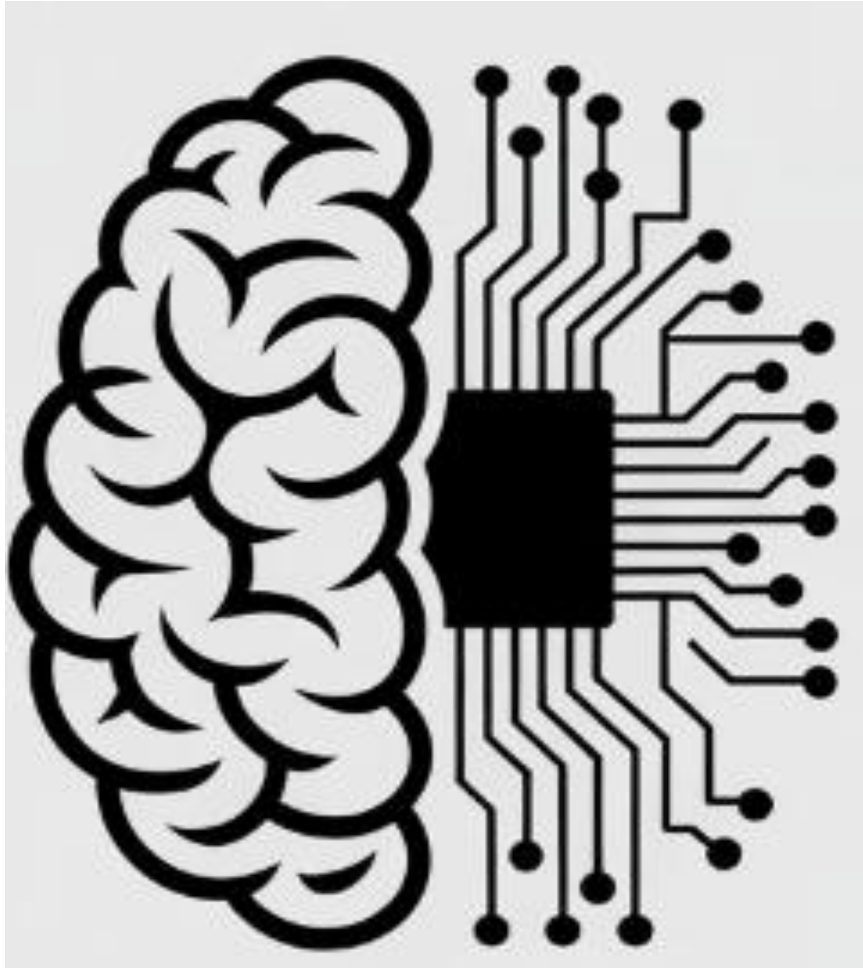

Performance Measures

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Confusion Matrix

- A **confusion matrix** is a NxN table used to evaluate the performance of a classification model by comparing **actual** and **predicted class** labels.

		True Class	
		Positive	**Negative**
Predicted Class	**Positive**	**True Positive (TP)**	**False Positive (FP)**
	Negative	**False Negative (FN)**	**True Negative (TN)**

For a binary classification problem, we would have a 2 x 2 matrix as shown in figure with 4 values.

Evaluation: Confusion Matrix

TP – True Positive: Number of correct positive predictions that are actually positive.

FP – False Positive: Number of incorrect positive predictions that are actually negative.

FN – False Negative: Number of incorrect negative predictions that are actually positive

TN – True Negative: Number of correct negative predictions that are actually negative.

		True Class	
		Positive	Negative
Predicted Class	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Evaluation Metric

1) Accuracy

- It measures how many **predictions** are **correct** out of total predictions.
- It shows overall performance of the model.
- Suitable when classes are balanced.

$$\text{Accuracy} = \frac{TP + TN}{TP+FN+FP+TN} \qquad \text{Misclassification Error} = \frac{FP + FN}{TP+FN+FP+TN}$$

- Overall correctness of the model.

Evaluation Metric

2) Precision

- Precision measures how many **predicted positive** cases are **actually positive**.
- It focuses on reducing **false positives**.

$$\text{Precision} = \frac{TP}{TP+FP}$$

- Reliability of **positive predictions**.

Evaluation Metric

3) Recall

- It measures how many **actual positive** cases are correctly detected.
- It focuses on reducing **false negatives**.

$$\text{Recall} = \frac{TP}{TP+FN}$$

- Ability to detect **positive cases**.

Evaluation Metric

4) F1-score

- It is the **harmonic mean** of **precision** and **recall.**
- It **balances** both **precision and recall**.
- Useful when dataset is **imbalanced**.
- F1 = 2 × (Precision × Recall) / (Precision + Recall)

$$F1 - \mathrm{Score} = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

- **Balanced performance** measure.

Evaluation Metric-Summary

1. **Accuracy:** Overall correct predictions

2. **Precision:** Correctness of positive predictions

3. **Recall:** Detection of actual positives

4. **F1-score:** Balance between precision and recall

Exercise

- A medical test is used to detect a disease. The following results were obtained:
  - 40 **patients actually** had the disease and were correctly **predicted as positive**
  - 10 patients **had the disease** but were **predicted negative**
  - 30 patients **did not have the disease** and were correctly **predicted negative**
  - 20 patients **did not have the disease** but were **predicted positive**
- **Construct the confusion matrix**
- Compute:
  - Accuracy
  - Precision
  - Recall
  - F1-score

TP = 40
FN = 10
TN = 30
FP = 20

		True Class	
		Positive	**Negative**
Predicted Class	**Positive**	40	20
	Negative	10	30

Exercise

TP = 40
FN = 10
TN = 30
FP = 20

- **Accuracy**

$$Accuracy = \frac{TP + TN}{Total}$$

$$= \frac{40 + 30}{100} = 0.70$$

- Accuracy = 70%

Model is reasonably accurate but not highly reliable.

$$\text{Misclassification Error} = \frac{FP \quad +FN}{TP+FN+FP+TN}$$

Misclassification Error = 1 – accuracy
= 1 – 0.70 = 0.30 or 30%

		True Class: Positive	True Class: Negative
Predicted Class	Positive	40	20
	Negative	10	30

Exercise

TP = 40
FN = 10
TN = 30
FP = 20

- **Precision**

Precision is moderate and should be improved.

$$Precision = \frac{TP}{TP + FP}$$

$$= \frac{40}{40 + 20} = 0.67$$

- Precision = 66.7%

		True Class: Positive	True Class: Negative
Predicted Class	**Positive**	**40**	**20**
	Negative	**10**	**30**

Exercise

TP = 40
FN = 10
TN = 30
FP = 20

- **Recall**

Recall is strong and desirable.

$$Recall = \frac{TP}{TP + FN}$$

$$= \frac{40}{40 + 10} = 0.80$$

- Recall = 80%

Predicted Class	True Class: Positive	True Class: Negative
Positive	40	20
Negativ e	10	30

Exercise

TP = 40
FN = 10
TN = 30
FP = 20

- **F1-score**

Balanced but slightly biased toward recall.

$$F1 = \frac{2PR}{P + R}$$

$$= \frac{2(0.67)(0.80)}{0.67 + 0.80} = 0.73$$

- F1-score = 73%

True Class

Predicted Class		Positive	Negative
	Positive	40	20
	Negative	10	30

Analysis

$$Accuracy = \frac{TP + TN}{Total}$$

For Imbalanced Dataset: Classification Accuracy is not a correct measure

- **1. Accuracy = 70%**
  - **What it tells us:**
    - Accuracy shows the **overall correctness** of the model.
    - It answers: “**Out of all patients, how many were correctly classified**?”
  - **Analysis:**
    - The model correctly **predicts 70 out of 100 patients**.
    - While this seems reasonable, accuracy alone can be misleading, especially in medical problems.
    - It does not distinguish between types of errors (false positives vs false negatives).

Analysis

$$Precision = \frac{TP}{TP + FP}$$

Do not rely on precision when recall is critical

- **2. Precision = 66.7%**
  - **What it tells us:**
    - Precision measures how **reliable positive predictions** are.
    - It answers: “When the model says a patient has the disease, how often is it correct?”
  - **Analysis:**
    - Out of all predicted positive cases (60), only 40 are truly diseased.
    - About 1 in 3 positive predictions is wrong (false alarms).
    - This indicates moderate reliability of positive predictions.

Analysis

$$Recall = \frac{TP}{TP + FN}$$

Avoid recall alone when false positives matter

- **3. Recall (Sensitivity) = 80%**
  - **What it tells us:**
    - Recall measures the model's ability to detect actual disease cases.
    - It answers: "Out of all patients who truly have the disease, how many did we correctly identify?"
  - **Analysis:**
    - The model correctly identifies 80% of diseased patients.
    - Only 20% (10 patients) are missed.
    - This is quite good, especially in medical diagnosis where missing a disease is risky.

Analysis

$$F1-\text{Score} = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

Avoid F1 when both classes (positive & negative) are equally important

- **3. 4. F1-Score = 72.7%**
  - **What it tells us:**
    - F1-score is the balance between precision and recall.
    - It is useful when we want a single measure of performance, especially with uneven errors.
  - **Analysis:**
    - The value (≈73%) shows a moderate balance:
    - Good recall (80%)
    - Lower precision (66.7%)
    - Indicates the model is better at detecting disease than avoiding false alarms.

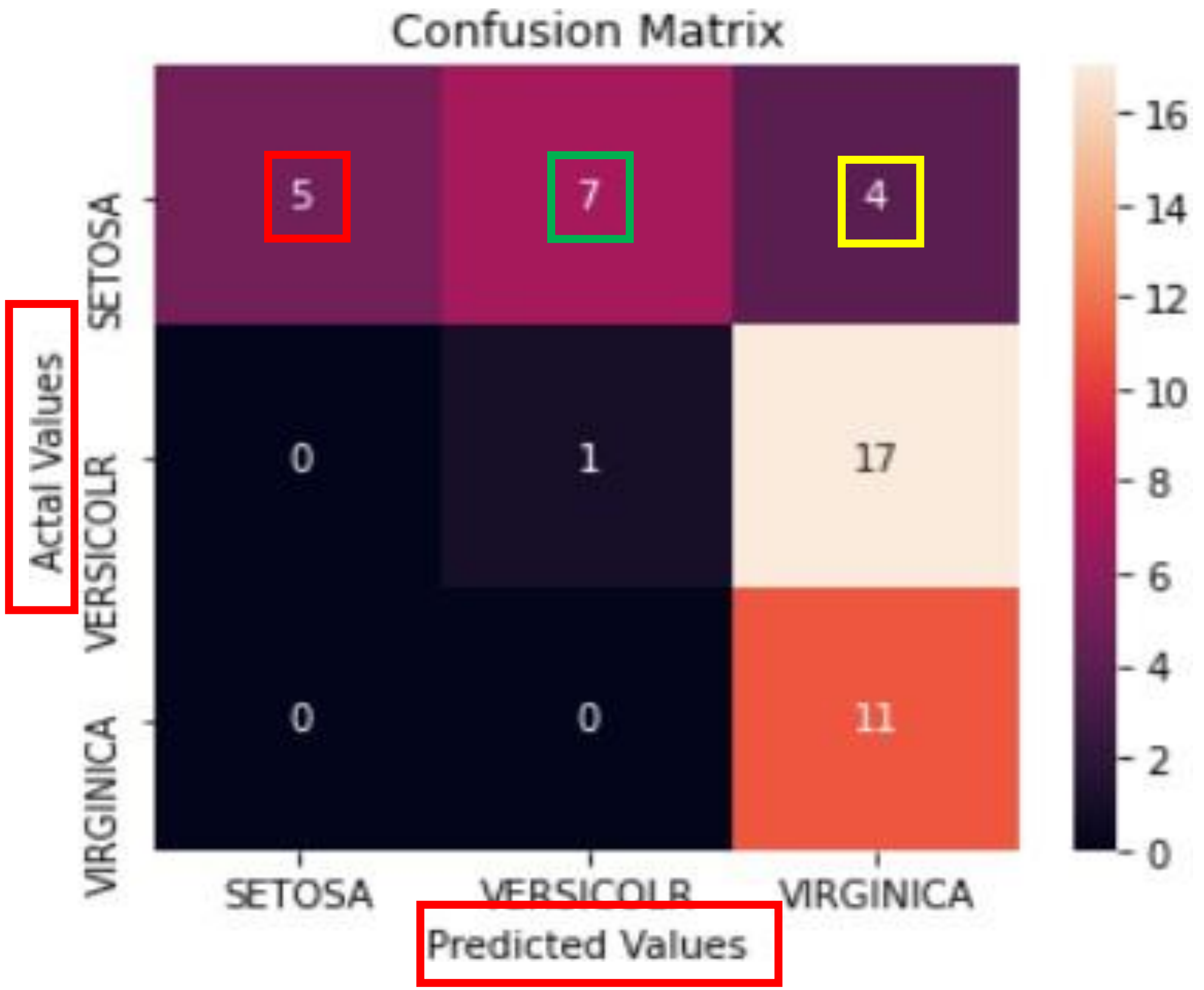
Confusion Matrix: Multiclass

- **IRIS Dataset**
  - The dataset has 3 flowers as outputs or classes,
    - **Versicolor**
    - **Virginica**
    - **Setosa.**

  - How to calculate TP, FP, TN, FN

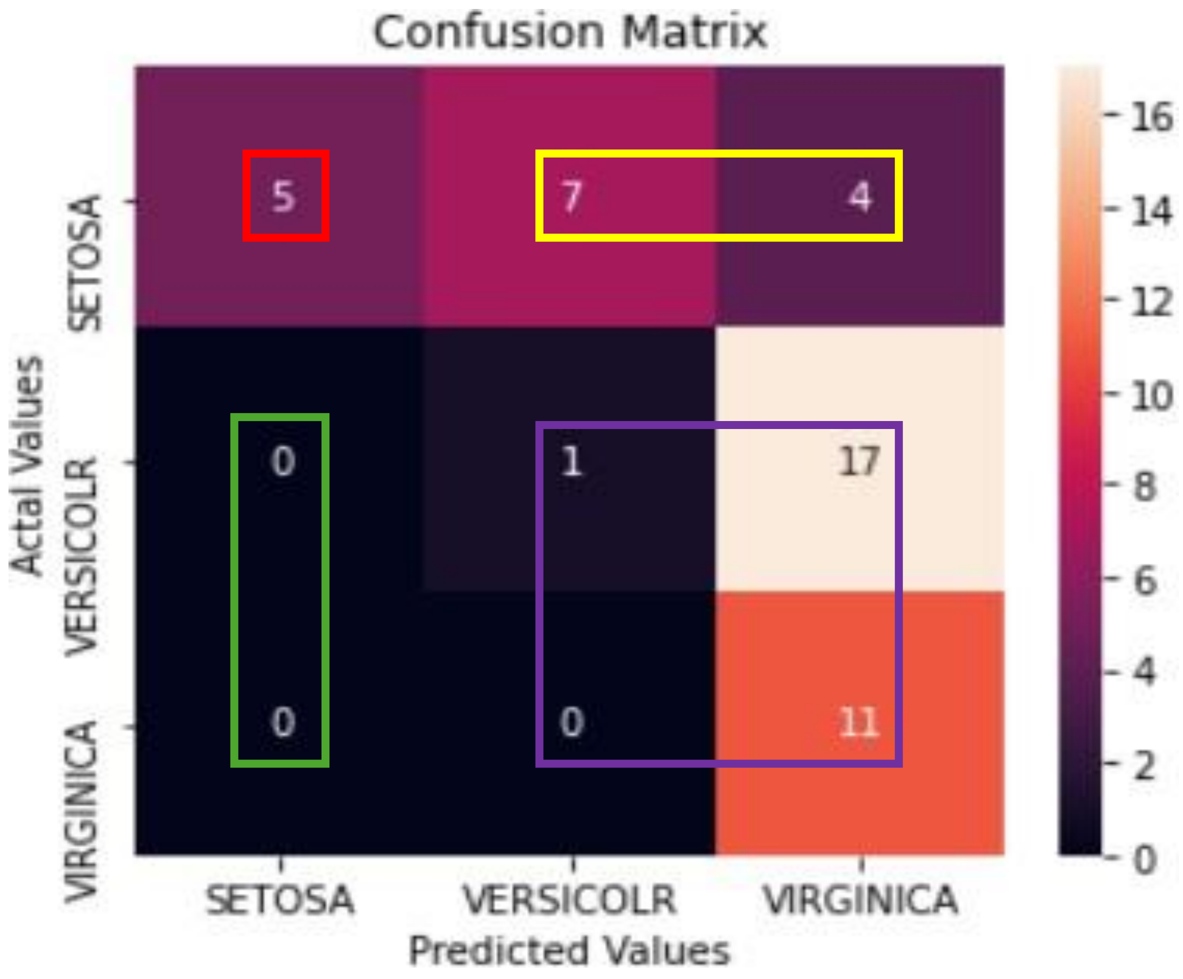
Confusion Matrix: Multiclass

- **What it tells?**
- **Actual**
  - **Setosa 5+7+4 = 16**
  - **Versicolr = 0+1+17 = 18**
  - **Virginica = 0+0+11 = 11**
- **Predicted**
  - **Out of the 16 Setosa**
    - **5 are predicted correct**
    - **7 are predicted as vericolr**
    - **4 are predicted as virginica**



Confusion Matrix: Multiclass

- **How to calculate FN, FP, TN, TP?**
- **SETOSA:**
  - **TP:** **5**
  - **FN:** **7 + 4**
  - **FP:** **0 + 0**
  - **TN:** **1 + 17 + 0 + 11**



H.W: compute for others as well

Confusion Matrix: Multiclass

- **How to calculate FN, FP, TN, TP :**
- **FN:** The False-negative value for a class will be the sum of values of corresponding rows except for the TP value.
- **FP:** The False-positive value for a class will be the sum of values of the corresponding column except for the TP value.
- **TN:** The True Negative value for a class will be the sum of values of all columns and rows except the values of that class that we are calculating the values for.
- **TP:** The True positive value is where the actual value and predicted value are the same.

Confusion Matrix: Multiclass

- **MNIST Dataset: labels[0-9]**

```
[[5825    1   37    9   12   20   31   13   12   22]
 [   0 6657   53   23   14   34   25   46  101   17]
 [   1   19 5627   44   12   13    6   52   12    3]
 [   4   21   50 5827    0   67    2   24  111   59]
 [   6   12   48    6 5480   33    8   40   33   61]
 [  10    7    9   79    1 5095   38    2   13    8]
 [  26    1   30   14   43   45 5789    4   20    2]
 [   1    4   29   34    6    1    0 5872    3   26]
 [  38   12   63   51   10   51   19   11 5447   24]
 [  12    8   12   44  264   62    0  201   99 5727]]
```

Micro and Macro Averaging (in classification evaluation)

- When you have multi-class classification, metrics like **precision, recall, and F1-score** can be computed in different ways. The two most important strategies are:
  - Macro averaging
  - Micro averaging

1. Macro Averaging (Class-wise Equal Importance)

- Compute the metric individually for each class, then take the simple average.

For each class i:

1. Compute Precision_i, Recall_i, F1_i
2. Then average:

$$\text{Macro Precision} = \frac{1}{K}\sum_{i=1}^{K} \text{Precision}_i$$

(where K = number of classes)

When to use

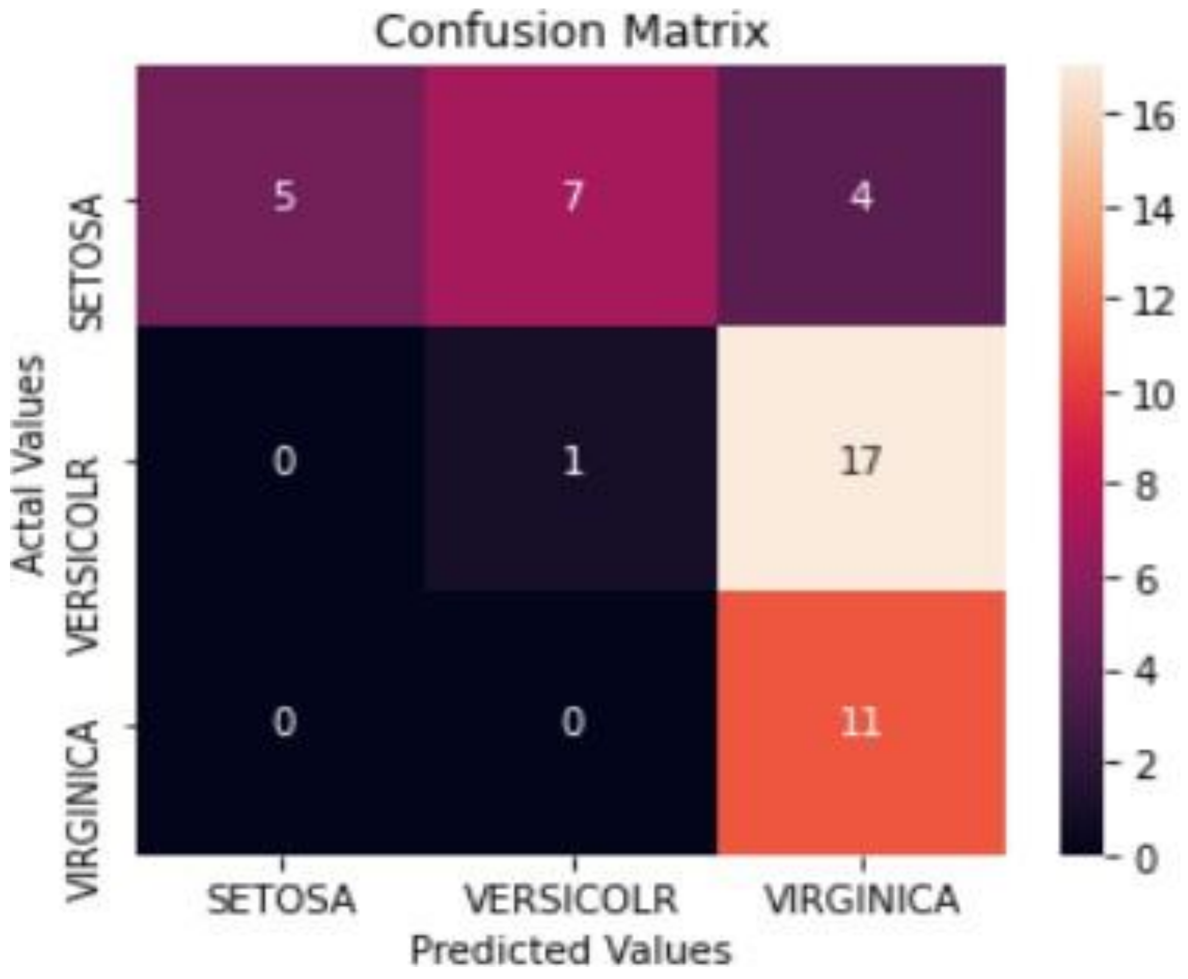
- Classes are imbalanced
- You care about minority class performance
- You want a fair evaluation across all classes

If:
Class A = very large (95% data)
Class B = very small (5% data)
Macro ensures Class B is **not ignored**

Macro Averaging

- **Step 1: Total samples**
  - **N = 5+7+4+0+1+17+0+0+11 = 45**

- **Step 2: Compute**
  - **TP, FP, FN per class**



2. Micro Averaging (Global Performance)

- Combine contributions of all classes first, then compute the metric globally.

Use it when:

- Dataset is balanced or naturally distributed
- You care about overall system accuracy
- You want performance at the instance level

1. Sum all:
   - True Positives (TP)
   - False Positives (FP)
   - False Negatives (FN)
2. Then compute:

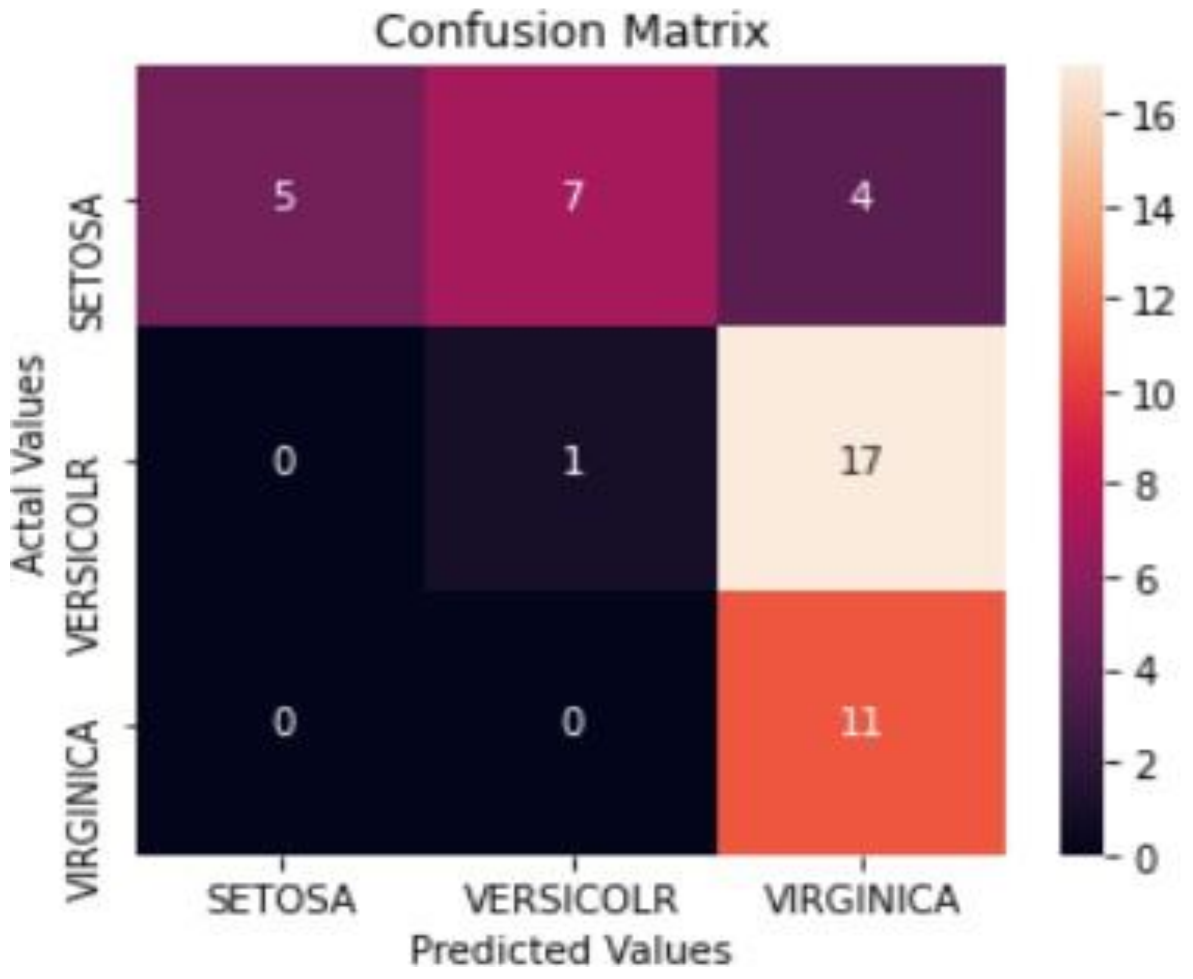
$$\text{Micro Precision} = \frac{\sum TP}{\sum TP + \sum FP}$$

$$\text{Micro Recall} = \frac{\sum TP}{\sum TP + \sum FN}$$

Micro Averaging

- **Step 1: Total samples**
  - **N = 5+7+4+0+1+17+0+0+11 = 45**

- **Step 2: Compute Total**
  - **TP, FP, FN**



Micro and Macro Averaging (in classification evaluation)

- Macro = “How well does the model perform on each class fairly?”
- Micro = “How many predictions are correct overall?”